

National Exam of Higher National Diploma 2026 Session

Specialty/option: SOFTWARE ENGINEERING (SWE)

Paper: Systems Analysis and Design

Duration: 4 hours

Credits: 6

GENERAL INSTRUCTIONS.

- **ANSWER ALL QUESTIONS**
- *You are reminded of the necessity of good English and orderly presentation of your material.*

SECTION A: MULTIPLE CHOICE QUESTIONS (50 MARKS)

Instructions: This paper is made up of three sections from which you are to *answer all the questions* in each section.

SECTION A: INFORMATION SYSTEMS AND DATABASE (25 marks)

Part I: Object Modelling (UML) (25 Marks)

The Ministry of Secondary Education in Cameroon wants to create CamEduNet, a computer system to store students' examination records. The system will keep information about:

- Students
- Exam Centres (e.g., Buea, Douala, Yaoundé)
- Subjects (Maths, English, ICT, etc.)
- Exam Sessions (2023, 2024, etc.)
- Results
- Teachers who upload results

A student can take many subjects. Each subject belongs to an exam session. Teachers enter results into the system.

Question 1: (8 Marks)

- A) What does UML stand for? (2 marks)
- B) In simple terms, explain why UML is useful when designing a system like CamEduNet. (2 marks)
- c) List four UML diagrams and state whether each one is structural or behavioural. (4 marks)

Question 2: (12 Marks)

Draw a simple UML class diagram for CamEduNet using these classes:

- Student
- ExamCentre
- Subject
- ExamSession
- Result
- Teacher

Your diagram should show:

- A) At least two attributes for each class (e.g., Student: studentID, name). (4 marks)
- B) Show relationships between the classes with multiplicity (1, *, etc.). (4 marks)
- C) Show one example of composition or aggregation and explain in one sentence why you chose it. (2 marks)
- D) Show one example of generalization (inheritance), for example: Student → Internal Student, External Student. (2 marks)

Question 3: (5 Marks)

- A) Draw a simple use case diagram for "Manage Examination Results" with at least three actors (Teacher, Student, Admin). (3 marks)
- B) Draw a simple sequence diagram for the use case "Upload Results". Show: Teacher → System → Database. (2 marks)

Part I: Database Administration With MYSQL (25 Marks)

Question 4: MCQ One Mark (01) For Each Correct Answer (20 Marks)

1. Which of the following is NOT a fundamental objective of a database?
 - A. Data redundancy
 - B. Data integrity
 - C. Data security
 - D. Data consistency
2. In the term ACID, the letter 'A' stands for:
 - A. Accuracy
 - B. Availability
 - C. Atomicity
 - D. Authorization
3. Which MySQL statement is used to create a new database?

- A. MAKE DATABASE EduResultsDB;
- B. CREATE DATABASE EduResultsDB;

- C. NEW DATABASE EduResultsDB;
- D. ADD DATABASE EduResultsDB;

4. Which SQL keyword is used to define a primary key?
 - A. UNIQUE

- B. KEY
 - C. PRIMARY KEY
 - D. MAIN KEY
5. In MySQL, **AUTO_INCREMENT** is used for:
- A. Automatically deleting records
 - B. Automatically increasing numeric values
 - C. Automatically updating tables
 - D. Automatically backing up data
6. Which SQL command is used to insert data into a table?
- A. ADD
 - B. INSERT INTO
 - C. PUT INTO
 - D. SAVE INTO
7. Which command displays all records from a table named students?
- A. SHOW students;
 - B. SELECT * FROM students;
 - C. DISPLAY students;
 - D. GET students;
8. Which SQL command starts a transaction?
- A. START
 - B. BEGIN
 - C. RUN
 - D. EXECUTE
9. If an error occurs in a transaction, which command cancels all changes?
- A. CANCEL
 - B. DELETE
 - C. ROLLBACK
 - D. RESET
10. Which command permanently saves changes in a transaction?
- A. SAVE
 - B. COMMIT
 - C. STORE
 - D. FINISH
11. Which MySQL command creates a new user?
- A. ADD USER
 - B. CREATE USER
 - C. MAKE USER
 - D. NEW USER
12. Which command is used to give a user permissions in MySQL?
- A. ASSIGN
 - B. GRANT
 - C. PERMIT
 - D. ALLOW

13. Which tool is commonly used to back up a MySQL database?
- A. mysqlbackup
 - B. mysqldump
 - C. mysqlcopy
 - D. mysqlsave
14. Which SQL clause is used to filter records?
- A. FILTER
 - B. WHERE
 - C. SELECT
 - D. FROM
15. Which column is best to index for searching students by surname?
- A. student_id
 - B. surname
 - C. gender
 - D. exam_centre_id
16. What does FK stand for in database design?
- A. File Key
 - B. Foreign Key
 - C. Final Key
 - D. Fixed Key
17. A foreign key in one table refers to:
- A. Any column
 - B. A primary key in another table
 - C. A text column only
 - D. A duplicate column
18. Which data type is best for storing gender (M/F)?
- A. INT
 - B. ENUM
 - C. DATE
 - D. FLOAT
19. Which command removes all data from a table but keeps its structure?
- A. DROP
 - B. DELETE
 - C. TRUNCATE
 - D. REMOVE
20. Which statement correctly creates a simple user?
- A. CREATE USER edu_admin IDENTIFIED BY 'Edu1234';
 - B. NEW USER edu_admin 'Edu1234';
 - C. ADD USER edu_admin PASSWORD 'Edu1234';
 - D. MAKE USER edu_admin 'Edu1234';

Question 5: (5 Marks)

- A) Write SQL to create a database named EduResultsDB.
- B) Write SQL to create a table student with:

(2 marks)

- i. Student_Id (Primary Key, Auto Increment)
- ii. Surname
- iii. Other_Names
- iv. Gender

(1+0.5+1+0.5 = 3 marks)

SECTION B: Web and Mobile programming

50 MARKS

Part I: Web programming (25 Marks)

Question 6: (5 Marks)

- A) What are tags? Give two HTML tags that don't come in a pair.. (3 marks)
- B) What are style sheets? Explain three ways of applying style sheets in a web document. (3 marks)
- C) What difference do you make between the followings
 - i. intranet and extranet? : (2 marks)
 - ii. Web page and web document? (2 marks)
- D) Define Internet and state four of its services. (4 marks)
- E) What are some of the common lists that can be used when designing a page? (2 marks)
- F) Create an HTML5 registration form with the following input fields for:
 - i. Full Name,
 - ii. Email,
 - iii. Phone Number,
 - iv. Date of Birth
 - v. A dropdown select for Blood Group
 - vi. A submit button

(6 × 0.5 = 3 marks)
- G) Link to an external CSS file named 'styles.css'. (2 marks)
- H) Write CSS code to style the form with a centered layout and border. (2 marks)
- I) Write a JavaScript function validateEmail() that checks if an email address contains the '@' symbol. (2 marks)

Part II: Mobile programming (25 Marks)

Question 7 (10 Marks)

MCQ One Mark (01) For Each Correct Answer

- 1) The keyboard, mouse, display, and system unit are:
 - A. hardware
 - B. output devices
 - C. storage devices
 - D. software
- 2) Programs that coordinate computer resources, provide an interface, and run applications are known as:
 - A. application programs
 - B. operating systems
 - C. storage systems

- D. utility programs
- 3) A browser is an example of a:
- A. general-purpose application
 - B. specialized program
 - C. system application
 - D. utility program
- 4) Although not as powerful as a supercomputer, this type of computer is capable of great processing speeds and data storage.
- A. mainframe
 - B. midrange
 - C. laptop
 - D. tablet
- 5) Apple's Watch is what type of computer?
- A. laptop
 - B. smartphone
 - C. tablet
 - D. wearable
- 6) RAM is a type of:
- A. computer
 - B. memory
 - C. network
 - D. secondary storage
- 7) Unlike memory, this type of storage holds data and programs even after electric power to the computer system has been turned off.
- A. primary
 - B. RAM
 - C. ROM
 - D. secondary
- 8) The type of file created by word processors, for example, memos, term papers, and letters.
- A. database
 - B. document
 - C. presentation
 - D. worksheet
- 9) Uses the Internet and the web to shift many computer activities from a user's computer to computers on the Internet.
- A. cloud computing
 - B. high definition
 - C. network
 - D. solid-state storage
- 10) The largest network in the world is [the]:
- A. Facebook
 - B. Internet
 - C. supercomputer

D. web

Question 8 (5 Marks)

A mobile application can be native or hybrid. With examples each explain what you understand by:

A) Hybrid mobile application. (3 marks)

B) Native mobile application. (2 marks)

Question 9: (10 marks)

As the number of smartphone and tablet devices increases, so does the risk that hackers and computer criminals will target these devices. Since they were first introduced, certain mobile devices have been vulnerable to eavesdropping and fraudulent charges. In addition, many of them automatically track a user's location. The major security risks associated with mobile devices include: *Malware*—Android apps, in particular, are vulnerable to malware because of the platform's openness.

To guard against malware threats, users have to upload the most recent versions of the operating system and use mobile security tools. *Premium SMS billing*—With their devices' vulnerability to malware, smartphone users face an added risk of subscribing to premium text-messaging services that charge every time users interact with them. Most cell phone carriers allow subscribers to block premium SMS messaging, however. *E-mail and SMS phishing*—Because it is more difficult to establish a link's legitimacy on a mobile device, mobile users are more likely to click on them, which is a phisher's dream come true. Mobile users should therefore use a lot of caution when using e-mail on these devices. *Spyware*—Commercially available software can be used by intruders to track and control the user's mobile activities. *Malicious Web sites*—These could pose a threat in the future, given that many smartphone browsers are based on a browser engine with vulnerabilities. Never leave your mobile device unlocked. Never leave it unattended. Always protect it with a password.

Work required

A) Give three (03) examples of security risks associated with a mobile device? (3 marks)

B) How can these devices automatically track a user's location? (3 marks)

C) Give four (04) recommendations for protecting your mobile device against these threats?

(4 marks)